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Product Design Management of Heritage Batik: Integrating Symbolic and Aesthetic Value for Generation Z Market Adoption

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Abstract: The sustainability of traditional cultural products relies on the capacity of producers to transform consumers' preferences into product design and marketing approaches. In the batik micro, small, and medium enterprises (MSME) context, Generation Z (Gen Z) is one of the key emerging market segments whose preferences depend not only on cultural meanings but also on symbolic value, aesthetics, usability, and relevance of the product. Despite the fact that the Theory of Planned Behaviour (TPB) is extensively applied in explaining purchase intention, only few researchers have attempted to apply this theory to help make product design and management decisions in heritage-based MSMEs. Thus, this study is intended to investigate the impact of symbolic value and aesthetic value on Gen Z's purchase intention towards traditional batik using an extended TPB model. A quantitative research approach was chosen and 208 participants from Gen Z were surveyed in three cities, namely Yogyakarta, Palembang, and Makassar. Structural equation modelling (SEM) was applied to assess the relationships. The results reveal that symbolic value has significant positive effects on attitude (Estimate = 3.077, p = 0.018), subjective norms (Estimate = 2.046, p = 0.017), and perceived behavioural control (Estimate = 1.677, $p < 0.001$). On the other hand, aesthetic value don't significantly affect subjective norms (Estimate = -1.348, p = 0.104) and perceived behavioural control (Estimate = -0.994, p = 0.011), but it have a significant negative effect on attitude (Estimate = -2.312, p = 0.068). Attitude and behavioural control significantly influence purchase intention, whereas subjective norms do not. Purchase intention has a significant positive effect on behaviour. The model demonstrates explanatory power, with R^2 values of 0.72 for purchase intention and 0.70 for attitude. These findings contribute to engineering management by showing how SEM-based consumer insights as a decision-support can guide batik MSMEs in product design, product-line segmentation, pricing accessibility, and youth-oriented market adoption..

Keywords: Product management; Cultural innovation; Symbolic value; Aesthetic value; Theory of Planned Behaviour; Generation Z; Consumer behaviour

1 Introduction

Growing interest in cultural innovation has strengthened interest in sustainability, market revitalization, and product management within the heritage industry in modern markets. Cultural fabrics, especially batik, are more than just cultural artifacts or fashion goods. They are also heritage products which include elements of design, symbolic meaning, representation, production, relevance to the market, and experience for the consumer.

Cultural heritage goods are characterized not only by practical characteristics but also by symbolic value, aesthetics and sociocultural stories stemming from tradition and culture [1, 2]. At the same time, in an era of globalization, e-commerce and fast-paced changes in the fashion industry, heritage goods face challenges related to their continued relevance, especially in the case of Generation Z (Gen Z) consumers [2–4]. Batik is one of the cultural heritage products that has important value for the Indonesian people. Batik does not only function as a fashion product. Batik also reflects cultural identity, social values, and traditions that are passed down from generation to generation [1, 2]. The existence of batik shows that a product can have economic value as well as cultural value. In addition to having a practical function, batik holds various symbolic and aesthetic meanings. Each motif contains a different cultural message. The colors and shapes used also reflect certain characters and philosophies. These values make batik different from fashion products in general [1, 2].

Changes in the business environment present new challenges for cultural heritage products. Globalization intensifies competition between products. Digital commerce is changing the pattern of interaction between producers and consumers. At the same time, the fast fashion industry continues to generate new trends in a short period of time [2–4]. Such shifts influence the selection and usage behaviour of consumers. The effect of the above shifts is evident in Gen Z population. They get information via social media and other online platforms.

When it comes to traditional batik micro, small, and medium enterprises (MSMEs), these challenges go beyond producer awareness of consumer preferences. In addition to that, they include the need to manage, redesign, position, and distribute cultural products in order to compete in the modern fashion industry. As one of the major consumer groups with continuously increasing buying capacity, Gen Z examines the products both in terms of their functionality and symbolism, aesthetics, affordability, usability, and applicability to daily life situations. It means that the batik MSMEs need to transform the cultural value and aesthetic identity of their products into more manageable product-specific decisions such as motif creation, complexity of design, storytelling related to culture, product portfolio composition, pricing accessibility, simplicity of distribution channels, and product positioning. Thus, the sustainability of heritage fashion depends not only on cultural preservation, but also on producers' capability to incorporate cultural values into a more systematic and innovation-oriented product management approach [5]. In terms of theory, the Theory of Planned Behaviour (TPB) is one of the well-known theories used to explain consumer purchase intentions. The theory indicates that purchase intention depends on attitudes, subjective norms, and behavioural control [6]. Many studies have been conducted using the TPB in order to explain consumer behaviour in various situations. However, the majority of the research still considers the psychological and utilitarian side of consumption. The cultural value within the products is still considered less important in the majority of the studies. This issue is more apparent in research related to heritage products. In general, many studies focus primarily on the conservation of culture and the production of handicrafts [1, 2, 7]. Studies involving Gen Z and their interaction with heritage products are still limited [2].

This gap becomes important to understand from the point of view of engineering management as well as a systems-oriented approach in product management. Heritage products such as batik have interdependencies in the areas of design, production, culture, marketing, consumer experiences, and distribution systems. In this setting, symbolic and aesthetic values should not be viewed solely as psychological factors influencing purchase intention. This is because they can equally be seen as values related to the product itself and they include product design, aesthetics, culture, and marketability. Symbolic value relates to the cultural story and authenticity behind batik products, while the aesthetic value refers to visual design, colors, design composition, and suitability for use.

For a country like Indonesia, which is still undergoing development, this problem has been more evident because of its sociocultural diversity, its important place in MSMEs, and new trends in lifestyle. Indonesia is one of the world's most populous countries (278 million people) with an impressive cultural diversity (more than 1200 elements of intangible cultural heritage). This condition provides an opportunity to integrate traditional values with modern consumer culture, particularly for young consumers. The development of contemporary lifestyles has greatly influenced consumers' preferences, particularly for the younger generation. On the one hand, batik possesses strong cultural legitimacy and symbolic representation, but on the other hand, batik SMEs must continuously adapt the design of the product market, as well as product pricing, promotion, and distribution strategies, to ensure that batik products remain relevant, wearable, competitive, and also accessible to younger consumers.

To address this gap, this study applies an extended TPB by incorporating symbolic value and aesthetic value as antecedent factors that may influence attitude, perceived behavioural control, purchase intention, and batik-related behaviour among younger consumers. Rather than viewing heritage clothing only as an object of consumption, the study positions batik as a product-management system shaped by heritage value, product design, user perception, and market acceptance.

The theoretical contribution of this study lies not merely in the inclusion of symbolic and aesthetic values within the TPB framework. Rather, it lies in the repositioning of the broadened concept of TPB as a product management framework for the adoption of cultural heritage apparel. Under this approach, symbolic and aesthetic values are regarded not only as the psychological antecedents of an intention, but also are viewed as product-oriented components that may be controlled and may help guide the manifestation of cultural values and meaning, design simplicity, usability, differentiation of the portfolio and positioning on the market. In terms of practice, the above results give strategic recommendations for batik MSMEs, designers and policymakers for creation of culturally meaningful, aesthetically relevant, accessible and usable heritage fashion products for Gen Z's everyday use.

The remainder of this paper is organized as follows. Section 2 reviews the theoretical background related to consumer behaviour, cultural and symbolic values, product management, and heritage fashion. Section 3 describes the research methodology and the process used to develop and estimate the proposed model. Section 4 presents and discusses the empirical results, including hypothesis testing and their implications for engineering management and product innovation. Finally, Section 5 concludes the study, outlines its limitations, and proposes directions for future research.

2 Literature Review

2.1 Cultural Innovation and Heritage Fashion Consumption

Heritage Innovation is becoming a useful theory for understanding how heritage-based industries remain relevant in a changing marketplace environment. Different from the normal approach to product innovation which emphasises improvements in the functional characteristics of the product alone, heritage innovation makes use of the re-invention of heritage values, symbolism and aesthetic representations to make sure that cultural relevance rests on the principles of continuity and sustainability [8]. In relation to Engineering Management, heritage innovation may also be regarded as a product management process in that it demands that companies interpret the language of culture in their product design and production process, market positioning as well as market adoption. In heritage fashion, traditional fabrics like batik function not only as useful products but as cultural representations with an embedded history and symbolism in the society.

The art of batik is an example of multiple cultural representations in terms of design elements that include motifs, color schemes, and methods of production [9, 10]. Each motif reflects philosophical meanings, social structures, and spiritual values that have been transmitted across generations. The traditional batik designs used in Indonesia traditionally have connections with social status and philosophy. One example of a motif with such a connection is the Parang Rusak Barong, which has been identified as a motif for royalty and is found at the Yogyakarta Palace. The Parang Rusak Barong motif is one of the batik larangan, meaning that it has been associated with leadership, authority, and royalty. In another way, the Kawung motif has been linked to some philosophic values like discipline, unity with nature, and morality. In coastal regions, the batik motifs have been used to capture the cultural exchanges of different civilizations [11–13].

In today's market environment, the sustainability of conventional fashion items is determined by their ability to cope with changing customer tastes. Gen Z customers tend to appreciate fashion items which enable them to express themselves, identify their culture, and create aesthetic value as well as relevance. In relation to batik, Gen Z is more inclined to view batik fabrics as fashion as well as culture [14, 15]. It can be deduced from this that the act of consumption is not only influenced by the utility of the product, but also by its symbolic and experiential value. Thus, cultural innovation should be seen as vital for comprehending the ways in which traditional textiles may sustain their authenticity while adjusting to the new consumer demands. However, previous debates surrounding heritage fashion were often concerned about cultural preservation and sustainability, whereas cultural values and their interpretation in terms of product management strategies of SMEs were often neglected.

The development of batik into modern fashion products requires balancing both innovation and conservation. There are efforts to integrate the use of traditional designs with modern cuts and materials in order to improve the marketability of these designs without losing their heritage value [8]. The innovation process not only enhances aesthetic appeal but also improves the emotional relevance of these products among the younger generation. Therefore, cultural innovation plays a very important role in creating a link between the conservation of the heritage and competitiveness in the market. This research work is based on this concept where batik is viewed as a heritage fashion product whose adoption depends on cultural meaning, aesthetics, evaluations, and market relevance.

2.2 Theory of Planned Behaviour in Cultural and Fashion Contexts

The TPB is one of the most widely used theories to explain consumer behaviour. This theory was developed by Ajzen and Kruglanski [16]. TPB explains that a person's intentions are influenced by attitudes, subjective norms, and perceptions of behavioural control. Various studies have used TPB to explain consumer behaviour. The application of this theory can be found in studies on sustainable consumption, ethical fashion, and the purchase of cultural products [17, 18]. In the context of cultural heritage fashion, attitude play a very important role. Consumers value batik not only as a clothing product but also for the cultural significance embodied in it. Favourable attitudes are commonly shaped by perceived symbolic representation, visual appeal, and cultural value significance [19]. The higher the suitability, the more likely consumers are to have the intention to buy.

Subjective norms are related to the influence of the social environment on individual decisions. These influences can come from family, friends, or social communities [20]. In a society that is collective, social norms are often an important factor in the formation of behaviour. The use of batik in various official events shows the influence of social norms. People often use batik because the social environment considers the use of batik as something positive. In the context of batik, the perception of behavioural control can be related to price, product availability, design variation, and ease of obtaining products [18]. However, younger consumers could also portray more independent buying behaviour which will diminish the influence of peer pressure on the consumer's purchase intention.

Perceived behavioural control implies the evaluation of how difficult or easy it is for a person to perform certain aspects of behaviour, explaining issues like cost, availability, and accessibility of the product. Regarding batik, the availability of affordable and stylish design variety is instrumental in enhancing consumers' chances to buy it [18]. If consumers view batik as accessible and easy to wear, they tend to develop greater purchase intentions.

However, even though there are many positive aspects associated with the use of the TPB, there still exist certain limitations in applying this theory to heritage fashion products. In particular, most studies have applied TPB either to general fashion items or to sustainable consumption; however, the role of cultural values incorporated into heritage products has not been explored thoroughly. Batik has symbolic and aesthetic values which are not covered by the original TPB elements.

2.3 Symbolic Value and Consumer Behaviour Towards Heritage Products

Batik is just one instance of a product whose value can be defined as highly symbolic. Usually, such products are used as symbols of cultural identification and national pride. That means that the significance that batik symbolizes is much more valuable than its functional value. Symbolic value of batik is created due to different cultural elements that it includes. Every batik pattern symbolizes some message or philosophy. This message has been transmitted from generation to generation by means of culture [21, 22].

Besides its philosophical implication, batik can also be viewed as an instrument of social communication. In the past, some motifs were employed to denote someone's social status. Nowadays, batik is being used as a marker of cultural identity and patriotism [23]. The celebration of National Batik Day reinforces the symbolism of batik in society. The event proves that besides being regarded as a fashion item, batik is also part of the nation's identity.

When considering from the point of view of consumer behaviour, symbolic value plays an important role in forming attitudes and purchase intentions. Consumers are more likely to form positive attitudes towards those commodities which match their identity and values [24]. The above statement is particularly applicable to Gen Z whose consumer behaviour is associated with self-identity formation. Symbolic consumption enables individuals to communicate who they are, what they value, and how they relate to cultural heritage.

Various studies show that symbolic value has a positive influence on consumer attitudes and behaviour. This influence is seen more strongly in products that have a relationship with culture and identity [21, 22]. However, most studies still view symbolic value as a psychological factor. Discussions about the management of symbolic value as product attributes are still relatively limited. In fact, symbolic value can be translated into various product strategies. Cultural narratives can reinforce product differentiation. The authenticity of the motif can increase the perception of the value of cultural heritage. Brand communication can also strengthen the connection between products and consumer identities.

For batik MSMEs, symbolic value can be a source of excellence that is difficult for competitors to imitate. Therefore, this study does not only view symbolic value as a factor that influences attitudes and purchase intentions. Symbolic value is also seen as a manageable product attribute to support product development and marketing strategies.

2.4 Aesthetic Value in Heritage Fashion and Generation Z Preferences

Aesthetic value is one of the factors that often influence consumers' decisions in choosing fashion products. Consumers usually pay attention to visual appearance before considering other attributes. Therefore, visual appeal is an important part of the product evaluation process [25].

In batik products, aesthetic value is reflected through motifs, colors, and quality of workmanship. Each visual element gives it a different character. The combination of these elements forms a visual identity that distinguishes batik from other fashion products. Color plays an important role in building the aesthetic value of batik. Sogan colors are often associated with simplicity and serenity. Meanwhile, the color blue is often associated with peace and spirituality [26]. This meaning strengthens the relationship between the visual aspect and the cultural value contained in batik. The uniqueness of batik can also be seen from the production process. The written batik technique requires a high level of precision. The process requires a lot of skill and experience [27]. The end result often has a different character on each piece of fabric. This condition increases the perception of the uniqueness and authenticity of the product [15].

However, in a constantly evolving market, the aesthetic value of batik must also adapt. Members of Gen Z have grown up surrounded by an environment characterized by a lot of visual input and the frequent changes in aesthetic preferences associated with social media content. Hence, traditional designs that remain unchanged may become irrelevant regardless of their cultural significance.

Recent studies have proven that aesthetics have a great impact on consumers' perceptions, but whether this impact has any effect on buying behaviour depends on the intelligent blend of traditional authenticity and modern innovation [28]. Thus, there is a dilemma between preserving the original culture and becoming relevant to the market. It can be solved through design adaptation when using newer materials and modern shapes for batik.

It is clear that aesthetic value is a component that has a significant impact on consumer attitudes. Moreover, Gen Z is one of those categories of consumers who are influenced by aesthetic value in terms of their behaviour towards a brand or a certain product. This is why it would be reasonable to incorporate the component into the TPB framework as its inclusion allows for considering visual and experience-related aspects that are usually overlooked in previous research.

However, there are some assumptions in the literature that require close attention. First, aesthetic value is seen as something positive and straightforward that would always contribute to consumer attitude and purchase intentions. However, the case of batik shows that the assumption does not work for the product under analysis. Batik is associated with complexity, patterns, rich colors, and other factors that make the clothing look formal. It makes it impossible to wear batik every day for Gen Z consumers who prefer simplicity.

Previous studies indicate that aesthetics can affect consumer attitudes; however, its impact on purchase intention could be contingent upon the blend of cultural authenticity with design relevance in contemporary times [28]. There is a conflict between the uniqueness of culture and design relevance in the current time. Therefore, aesthetic value should be assessed in heritage fashion, not only in terms of aesthetic appearance but also as an element that could affect consumer perceptions regarding appropriateness and usability of the product.

Incorporation of aesthetic value into the extended TPB in this study helps to address the gap in the literature concerning the assessment of the impact of aesthetic value on attitude, behavioural control, and purchase intention of Gen Z consumers for batik.

2.5 Micro, Small, and Medium Enterprises Product Innovation and Market Management for Batik

MSMEs play an important role in maintaining the sustainability of the batik industry. Most batik production in Indonesia is still dominated by small- and medium-scale enterprises. This group not only contributes to the regional economy, but also plays a role in maintaining cultural preservation. However, batik MSMEs face various challenges. Market competition is getting tougher. Consumer preferences are changing increasingly rapidly. In addition, the development of digital technology is changing the way products are marketed and distributed.

Various studies have discussed consumer behaviour in the context of fashion. These studies cover conventional clothing, ethical fashion, sustainable fashion, second-hand clothing, and environmentally friendly fashion products [29, 30]. The results of the study provide an important understanding of the factors that affect consumer purchase intentions. Most of the research focused on explaining consumer behaviour. The focus is seen in studies conducted by Hosta and Žabkar [29], Sun et al. [7], and Magano et al. [31]. These studies explain how attitudes, norms, and values influence purchasing decisions. Research on sustainable fashion also shows a similar pattern. Various studies explain the influence of environmental awareness on consumer decisions [32–36]. However, attention to cultural aspects and product management is still relatively limited. Other studies addressed ethical fashion, second-hand clothing, modest activewear, and Gen Z fashion behaviours [30, 37, 38]. The results of the study show that identity, personal values, and social norms play a role in shaping consumer behaviour. Despite making important contributions, most research still stops at explaining purchasing behaviour. Existing studies has not explained much about how information about consumers can be used to support product management decisions.

In fact, batik MSMEs need more practical information. Business actors need to determine a design that suits market needs. They also need to manage product portfolios, set prices, and choose the right distribution channels. These challenges are becoming increasingly complex because batik MSMEs must manage various aspects simultaneously. Cultural values need to be maintained. The quality of the design needs to be improved. Prices must remain affordable. The product must also be easy for consumers to obtain. In such cases, the knowledge about the elements influencing product adoption is very crucial. The symbolic value may be applied in reinforcing cultural stories and authenticity of motifs. The aesthetic value may be applied in designing a product more appropriate for the demands of the market. Behavioural control perceptions may assist in understanding elements of price, availability, and usability.

In summary, the connection between consumer behaviour theory and product management is very applicable for batik MSMEs. This perspective enables companies to convert consumer preferences into sound business decisions. This study uses an extended TPB model to explain the relationship. Symbolic value and aesthetic value are positioned as factors that influence consumer attitudes, subjective norms, perception of behavioural control, intentions, and behaviour. Thus, this study not only explains consumer behaviour, but also provides a basis for decision-making in the management of batik products.

2.6 Research Gap and Hypothesis Development

Attention to cultural heritage fashion has continued to increase in recent years. Various studies have been conducted to understand consumer behaviour and cultural preservation efforts. However, a number of gaps can still be found in the existing literature. Most previous research has placed cultural heritage products in the context of cultural preservation and tourism. This approach makes an important contribution to cultural sustainability. However, this approach does not explain the decision-making process of consumers using an established behavioural framework.

Another limitation can be seen in the application of the TPB. This theory has been widely used to explain purchase intent. However, the integration of symbolic and aesthetic values into the TPB model is still relatively limited. This condition is especially seen in research that discusses traditional fabrics and cultural heritage fashion products.

The next gap is related to Gen Z. Research that specifically discusses the behaviour of this group towards cultural heritage products is still relatively limited. In fact, Gen Z is expected to become a very important market segment for the sustainability of the batik industry in the future.

From a product management perspective, the limitations of existing research are becoming increasingly apparent. Many studies explain why consumers appreciate a product. However, few studies have shown how the findings can be translated into product development strategies. Discussions on product portfolio management are still limited. Studies on design simplification are also still rare. The same condition can be seen in research on improving ease of use, pricing strategies, and product innovation for MSMEs.

Another gap arises in the discussion of aesthetic values. Most studies consider aesthetic value to always have a positive impact on consumer behaviour. In fact, visual characteristics that are too complex can cause barriers to use in certain consumer groups. In the context of batik, the complexity of the motif can increase the artistic value of the product. Nonetheless, this kind of complexity can make the product appear very formal. This situation can reduce young consumers' willingness to use batik products for their daily lives.

Another unconsidered aspect relates to the lack of knowledge about the way symbolic and aesthetic values work in the TPB paradigm. Symbolic value can increase attitude and behavioural control of consumers because it connects the product to consumers' identity and culture. At the same time, aesthetic value can affect consumers' evaluation of visual attractiveness, although its impact on heritage fashion can be indirect since visual complexity decreases the perceived ease of use of the product.

This research seeks to fill these gaps by developing an extended TPB model in which symbolic value and aesthetic value are treated as precursors of attitude, behavioural control, purchase intention, and behaviour towards batik products among Gen Z. The managerial interpretation of these findings is further discussed in the discussion section, particularly in relation to product management implications for batik MSMEs. The research gaps identified from the reviewed literature and the corresponding research contributions of this study are summarized in Table 1.

Table 1. Research gap matrix of previous studies

Ref.	Object	Population	A	SN	BC	I	B	ES	S	PN	EC	MM	PT	E
Hosta and Žabkar [29]	Clothing	Adult consumers aged 18–65 years	✓		✓			✓						
Liu et al. [30]	Ethnic fashion products	General consumers (China)	✓	✓	✓	✓		✓						
Sun et al. [7]	Clothing	General consumers (China)	✓	✓	✓	✓	✓	✓			✓			
Lee et al. [32]	Sustainable clothing	Consumers aged 12–65 years (Korea)	✓	✓	✓	✓	✓				✓			
Magano et al. [31]	Fashion products	Not specified	✓	✓	✓	✓	✓		✓					
Ong et al. [39]	Clothing	General consumers (Philippines)		✓	✓	✓	✓							
La Rosa and Jorgensen [33]	Sustainable apparel	General consumers (Turkey)	✓	✓	✓	✓	✓	✓						
Tiwari et al. [34]	Fashion products	General consumers (India)	✓	✓	✓	✓	✓						✓	
Hwang and Kim [38]	Modest activewear	Muslim women consumers (US)	✓	✓	✓	✓	✓		✓					
Nicolau et al. [40]	Sustainable fashion	General consumers (Brazilian)	✓	✓	✓	✓								✓
Sobuj et al. [41]	Apparel	General consumers (Bangladesh)		✓	✓	✓	✓				✓			
Vlastelica et al. [35]	Sustainable clothing	Gen Z consumers	✓	✓	✓	✓	✓				✓			
Liu et al. [36]	Green apparel	General consumers	✓	✓	✓	✓	✓							
This Paper	Batik	Gen Z (Indonesia)	✓	✓	✓	✓	✓	✓	✓					

Note: A = Attitude; SN = Subjective Norm; BC = Behavioural Control; I = Purchase Intention; B = Behaviour; ES = Aesthetic Value; S = Symbolic Value; PN = Personal Norm; EC = Environmental Concern; MM = Marketing Mix; PT = Perceived Trust; E = Eco-shame; Gen Z = Generation Z.

3 Methodology

In this study, an expanded version of the TPB model is used to understand why Gen Z adopts batik and to make relevant product management decisions for batik MSMEs as illustrated in Figure 1. The extended model includes Symbolic Value and Aesthetic Value as external variables that are relevant to heritage fashion products alongside the TPB variables. The constructs include Attitude, Subjective Norm, Behavioural Control, Intention, and Behaviour. Similar to the TPB, the constructs of Attitude, Subjective Norm, and Behavioural Control are considered as direct determinants of Intention, and Intention determines Behaviuor. The external variables of Symbolic Value and Aesthetic Value will influence the three antecedent constructs as the cultural meaning and Aesthetic Value affect how one evaluates the product, assesses its socially acceptable use, and feels capable of use batik. From the standpoint of engineering management, this model seeks to determine what Attributes should be managed regarding product development, cultural storyboarding, portfolio management, price determination, distribution channels, and positioning. The operational definition of constructs can be found in Table 2 and the proposed conceptual model in Figure 1.

Table 2. Operational definition of research variables

No.	Variables	Conceptual Philosophy	Operational Definition	Product Management Interpretation
1	Symbolic Value	An individual's perception of the cultural meaning, social identity, and symbolic value inherent in a product.	The value that individuals feel when batik is considered to represent identity, pride, and connection with Indonesian culture.	Cultural narrative strength, motif authenticity, heritage storytelling, local identity representation, symbolic branding, and cultural positioning.
2	Aesthetic Value	Perceived value is based on the visual beauty, design, and artistic appeal of a product.	Gen Z's perception of the beauty, color, and design of batik as a factor that influences their interest in wearing it.	Design complexity, color harmony, visual differentiation, contemporary styling, usability for daily wear, and compatibility with fashion trends.
3	Attitude	A person's positive or negative evaluation of certain behaviours.	Positive or negative assessment of the younger generation towards the use of batik as part of their lifestyle.	Consumer acceptance of batik product concepts, design direction, and market repositioning strategy.
4	Subjective Norm	An individual's perception of social pressures that encourage or inhibit certain behaviours.	The social influence that Gen Z feels from family, friends, or social media in their decision to wear batik.	Social endorsement, influencer-based promotion, community acceptance, digital campaign strategy, and social visibility of batik products.
5	Behavioural Control	An individual's perception of the ease or difficulty of performing certain behaviours.	Gen Z's perception of the ease of accessing, buying, and wearing batik on various occasions.	Price accessibility, product availability, channel convenience, size and style variety, daily wearing suitability, and omnichannel distribution.
6	Intention	A person's desire or plan to perform an action.	The level of Gen Z's desire to buy or wear batik in the near future.	Market demand indicator for product portfolio planning, product prioritization, and target segment strategy.
7	Behaviour	Real actions taken by individuals regarding certain behaviours.	The frequency and intensity of batik use by Gen Z in daily life or formal events.	Actual product adoption, repeat usage, occasion-based product development, and long-term market sustainability.

Note: Gen Z = Generation Z.

Based on this conceptual framework, the research hypothesis is formulated as follows:
H1. Symbolic Value has a positive effect on Attitude.
H2. Symbolic Value has a positive effect on Subjective Norm
H3. Symbolic Value has a positive effect on Behavioural Control
H4. Aesthetic Value has a positive effect on Attitude
H5. Aesthetic Value has a positive effect on Behavioural Control
H6. Aesthetic Value has a positive effect on Subjective Norm
H7. Subjective Norm has a positive effect on Intention
H8. Attitude has a positive effect on Intention
H9. Perceived Behavioural Control has a positive effect on Intention
H10. Intention has a positive effect on Behaviour

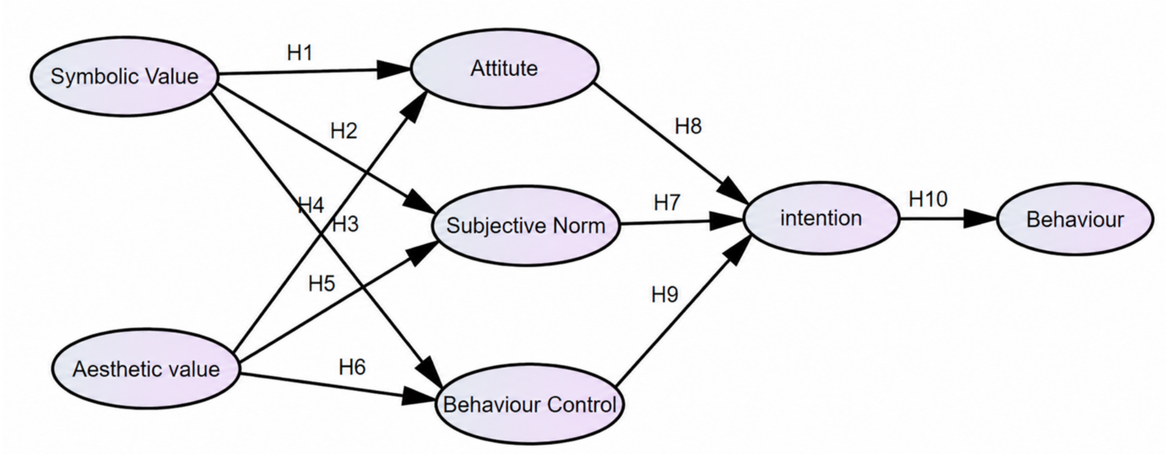

Figure 1. Proposed conceptual model and hypothesised relationships

3.1 Data Collection

Data were collected through a survey. The survey was used to obtain information about respondents' characteristics, consumer preferences, and factors that affect the intention to buy batik products. This research targeted consumers who have bought or used batik. The selected respondents were between 15 and 25 years old. Each respondent was also required to have purchased or used batik at least twice during the previous six months.

The research was conducted in three cities, namely Yogyakarta, Palembang, and Makassar. The three cities were chosen to represent different social and cultural contexts. Yogyakarta represents a region with a strong batik tradition. Palembang represents a non-Javanese urban market that has a Malay textile tradition. Meanwhile, Makassar represents an urban area in Eastern Indonesia with a strong influence on Bugis culture. The selection of the three cities aimed to obtain a variety of batik consumption contexts in young age groups. Therefore, this study is not intended to represent all Gen Z in Indonesia. This research focuses more on the diversity of batik consumption experiences in various regions.

Respondents were recruited through online questionnaires distributed through social media, university networks, and youth communities. Data collection was carried out in the period from January to March 2025. The purposive sampling technique was used because the young consumer population is very broad and diverse. This technique allows researchers to select respondents who have adequate experience and understanding of batik products. All data then went through a filtering process. Incomplete answers, duplicate data, and respondents who did not meet the study criteria were excluded from the analysis.

3.2 Structural Equation Modelling Analysis

Structural equation modelling (SEM) was employed to test the proposed conceptual model against the empirical data and to examine the relationships among the latent constructs simultaneously [42]. SEM consists of two main components: the measurement model and the structural model. The measurement model assesses the relationships between latent constructs and their observed indicators, whereas the structural model evaluates the hypothesised relationships among the latent constructs. In this study, SEM was used to assess construct validity, evaluate model fit, and test the proposed hypotheses.

4 Result and Discussion

4.1 Respondent Demographics and Product Purchasing Statistics

This study involved 208 respondents. This sample size was adequate for SEM analysis. Hair et al. [43] stated that a sample size of between 100 and 200 respondents is sufficient for an SEM model of moderate complexity. Bentler and Chou [44] also stated that an appropriate sample size ranges from five to ten times the number of research indicators. Since 27 variables were used in this research, the required sample size ranged from 135 to 270 respondents, while the sample size used lies within the stated range. As such, the sample size is sufficient to estimate the measurement model and the structural model. Nonetheless, because of the use of purposive sampling and only focusing on three cities, the sample size cannot be considered to be statistically representative of the whole Gen Z generation in Indonesia.

Table 3 shows the demographic profile of the respondents. Most of the respondents were male. This group accounted for 63.46% of the total respondents. Female respondents amounted to 36.54%. In terms of age, the age group of 20–25 years represented the largest age group. This group accounted for 67.79% of the total respondents. Respondents under the age of 20 accounted for 32.21%.

Table 3. Respondent demographics

Demographic Characteristic	Demographic Profile	Count	Percentage (%)
Gender	Male	132	63.46
	Female	76	36.54
Age (yr)	<20	67	32.21
	20–25	141	67.79
Income (IDR million)	<1	77	37.02
	1–3	99	47.60
	3–5	27	12.98
	>5	5	2.40
Ethnicity	Jawa	128	61.54
	Sunda	11	5.29
	Melayu	37	17.79
	Bugis	12	5.77
	Minangkabau	5	2.40
	Others	15	7.21
Total Sample		208	100

The age composition shows the dominance of young consumers. This age group has high exposure to fashion trends. This group is also active in using digital media. Their dressing preferences tend to change quickly. This condition is in accordance with the current development of batik. Batik is growing as an adaptive fashion product for the younger generation. Batik does not only function as a traditional garment. Batik is also a means to show identity and lifestyle.

Based on income level, most respondents are in the lower-middle-income group. As many as 47.60% of respondents have an income between IDR 1 million and IDR 3 million per month. As many as 37.02% of respondents have an income of less than IDR 1 million per month. Respondents with an income between IDR 3 million and IDR 5 million accounted for 12.98%. The group with an income above IDR 5 million only accounted for 2.40%.

The income structure describes the character of the batik market that is quite inclusive. Batik can reach various groups of people. This condition is especially seen among young consumers. Most young consumers have limited purchasing power. Despite this, they still show interest in products that have cultural value.

The findings show that batik is not always perceived as a premium product. Batik is also seen as an affordable product. Batik is also seen as a functional product. In addition, batik is considered relevant for use in daily activities.

These findings have immediate implications for product management. Batik MSMEs cannot rely only on premium products. MSMEs also cannot only focus on formal products. Business actors need to develop more affordable product lines. Simpler designs also need to be developed. Casual collections need to be expanded. This step can increase young consumers' access to batik products.

In terms of ethnicity, most of the respondents came from the Javanese tribe. This group accounted for 61.54% of the total respondents. Respondents from Malay ethnicity accounted for 17.79%. Respondents from the Bugis ethnic group accounted for 5.77%. Respondents from ethnic Sundanese accounted for 5.29%. Respondents from the Minangkabau ethnic group accounted for 2.40%. Other ethnic groups accounted for 7.21%.

The dominance of Javanese respondents is in line with the position of batik in Javanese culture. Batik has strong historical roots in the region. The use of batik is also wider than many other regions.

Nonetheless, the diversity of ethnic backgrounds shows interesting findings. Batik is no longer limited to a specific cultural identity. Batik has been accepted by various community groups. This condition shows the expansion of the function of batik in social life.

The findings show a shift in the meaning of batik. Batik is no longer just a symbol of local culture. Batik has evolved into a national fashion product. This product has Symbolic Value and Aesthetic Value that can be appreciated by various groups of people.

From the perspective of engineering management and cultural product innovation, cross-ethnic acceptance has an important meaning. Product development needs to maintain cultural authenticity. Product development also needs to pay attention to the needs of the wider market. MSMEs can use cultural narratives as a source of differentiation. At the same time, MSMEs need to adjust motifs, colors, models, and product formats to more diverse consumer characteristics.

4.2 Structural Model Evaluation

The analysed SEM model is the re-estimated final model developed after measurement model refinement (Figure 2). The initial model was first evaluated through confirmatory factor analysis (CFA), and two Symbolic Value indicators (S3 and S4) were removed because their loading values were below 0.50. The factors that had loadings equal to or more than 0.50 were retained in the measurement model. Therefore, the factor with a loading of exactly 0.50 was selected due to meeting the minimum threshold required. After the item deletion, both the CFA measurement model and the SEM structural model were re-estimated in AMOS 24 using the retained indicators. Therefore, all reported factor loadings, critical ratio, average variance extracted (AVE), variance inflation factor (VIF), model fit indices, path coefficients, R^2, and effect size values correspond to the final measurement and structural model. The re-estimated model integrates TPB with Symbolic Value and Aesthetic Value to explain the formation of Intention and Behaviour while also providing a managerial map for batik product management.

The removal of S3 and S4 was based on statistical criteria and theoretical consistency. These Symbolic Value attributes help to define the cultural heritage, feelings of pride, positive meaning within the community, and symbolism. This allows for the development of an understanding of Symbolic Value as a construct relating to products.

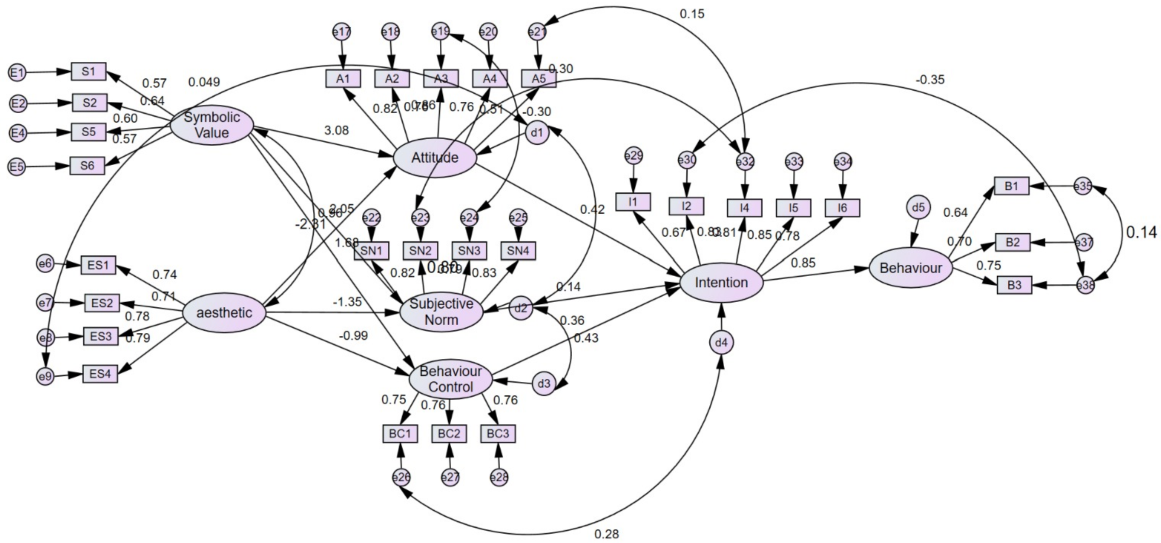

Figure 2. Result of structural equation modelling (SEM)

4.2.1 Confirmatory factor analysis

The CFA and the refinement of the proposed model were conducted using AMOS 24. As a result of the exclusion of S3 and S4, the measurement model became more stable regarding construct structure. Construct validity was estimated based on the factor loadings, composite reliability (CR), AVE, and VIF presented in Table 4. After re-estimation, the model became valid and demonstrated better construct stability. Construct reliability was evaluated by means of CR values presented in Table 4; all seven constructs had CR values ranging from 0.70 to 0.89, which means good internal consistency since, as a rule, CR values equal to 0.70 or higher are considered sufficient [42].

Besides the CR values, construct validity was evaluated by means of factor loadings of each indicator on the latent construct. All retained indicators had factor loadings equal to or greater than 0.50, satisfying the minimum recommended threshold for convergent validity. The obtained factor loading ranges for each construct were: Symbolic Value (0.58–0.64), Aesthetic Value (0.72–0.79), Attitude (0.75–0.85), Subjective Norm (0.79–0.82), Behavioural Control (0.75–0.77), Intention (0.66–0.85), and Behaviour (0.63–0.75). Thus, it can be concluded that all retained indicators adequately reflected their latent constructs; however, one should be careful while interpreting Symbolic Value construct since it has lower AVE than other constructs. The retained indicators demonstrated acceptable factor loadings, supporting sufficient convergent validity.

From the viewpoint of engineering management, it is important to note that the validity of constructs is critical for further development of research because each construct can be represented by some decision variables related to the product. Symbolic Value includes such constructs as cultural storytelling, motif authenticity, and identity-oriented positioning. Aesthetic value consists of design-related constructs such as color harmony, motif complexity, visual uniqueness, and modernity. Behavioural control refers to accessibility constructs including price, availability, channel convenience, and usability.

Table 4. Measurement items with factor loadings and construct validity

Construct	Indicator	Loadings	CR	AVE	VIF
	Symbolic Value		0.70	0.40	2.31
S1	Batik clothes are an important representation of my cultural heritage	0.58	–	–	–
S2	Wearing batik clothes evokes emotions such as pride, joy, or nostalgia in me	0.64	–	–	–
S5	My community views the use of traditional clothing in a positive way.	0.62	–	–	–
S6	Batik clothing has a significant symbolic meaning for the community	0.58	–	–	–
	Aesthetic Value		0.85	0.58	2.05
ES1	Batik clothes are aesthetically appealing	0.75	–	–	–
ES2	Colors, patterns, textures, and cultural meanings enhance the aesthetic value of Batik clothing	0.72	–	–	–
ES3	Originality and uniqueness are important to my aesthetic assessment of Batik clothing	0.79	–	–	–
ES4	The aesthetic appeal of batik clothes increases my confidence	0.79	–	–	–
	Attitude		0.86	0.56	3.19
A1	Wearing batik clothes makes me feel proud and comfortable.	0.83	–	–	–
A2	Batik clothes represent my personal identity.	0.75	–	–	–
A3	The batik clothes available to me feel authentic.	0.85	–	–	–
A4	My understanding of cultural heritage influenced the way I viewed Batik clothing	0.75	–	–	–
A5	I feel the need to use batik clothes	0.50	–	–	–
	Subjective Norm		0.88	0.65	3.12
SN1	There is pressure from the community in my community to conform to the norms of traditional dress.	0.82	–	–	–
SN2	I feel the need to adjust to my friends' preferences when choosing batik clothes	0.79	–	–	–
SN3	My family appreciates the wearing of batik clothes	0.79	–	–	–
SN4	My community strongly emphasizes the importance of Batik clothing	0.82	–	–	–
	Behavioural Control		0.80	0.57	2.03
BC1	I have enough financial resources to buy batik clothes	0.75	–	–	–
BC2	Batik clothing stores or platforms are easily accessible to me.	0.77	–	–	–
BC3	External factors (e.g., cultural appropriation, trends) affect my ability to wear batik clothes	0.75	–	–	–
	Intention		0.89	0.59	1.84
I1	I will likely buy cultural clothes in the next six months	0.66	–	–	–
I2	My personal interest in cultural heritage motivated me to buy cultural clothing	0.83	–	–	–
I4	The availability of batik clothes in stores encouraged me to make purchases.	0.81	–	–	–
I5	Cultural identity significantly influenced my intention to buy batik	0.85	–	–	–
I6	I find that some styles of batik clothing are suitable for modern settings	0.78	–	–	–
	Behaviour		0.82	0.50	1.84
B1	I have some batik clothes	0.63	–	–	–
B2	I often wear batik in my daily life.	0.70	–	–	–
B3	I feel comfortable wearing batik clothes	0.75	–	–	–

Note: S = Symbolic Value; ES = Aesthetic Value; A = Attitude; SN = Subjective Norm; BC = Behavioural Control; I = Purchase Intention; B = Behaviour; Loadings = Standardised Factor Loadings; CR = composite reliability; AVE = average variance extracted; VIF = variance inflation factor. The en dash (–) indicates that the corresponding value is not applicable because CR, AVE, and VIF were calculated at the construct level rather than for individual measurement items.

The AVE was generally above the minimum threshold of 0.50, meaning that most constructs sufficiently explained variance among their indicators. The construct called Symbolic Value had an AVE equal to 0.40, which is below the optimal threshold. Symbolic Value is one of the retained constructs as it has the factor loadings above 0.50 for all retained indicators, CR equal to 0.70, and has theoretical importance within the scope of heritage product management. It is done based on the statement of Fornell and Larcker [45] that convergent validity is still acceptable in case AVE is below 0.50 and CR is above 0.60. However, the low AVE is admitted as a methodological limitation, while interpreting Symbolic Value is oriented on the theoretical aspect of it.

Discriminant validity evaluation using the Fornell–Larcker approach showed that most constructs had AVE square root values greater than the correlations between the other constructs (Table 5). These results indicate that each construct is able to differentiate itself from the other constructs in the research model. Despite the conceptual closeness between Attitude and Subjective Norm constructs, this condition is still acceptable because the two constructs have different theoretical foundations within the TPB framework.

Table 5. Discriminant validity results based on the Fornell–Larcker criterion

Construct	S	ES	A	SN	BC	I	B
S	0.632	–	–	–	–	–	–
ES	0.657	0.762	–	–	–	–	–
A	0.575	0.526	0.748	–	–	–	–
SN	0.535	0.530	0.798	0.806	–	–	–
BC	0.461	0.518	0.654	0.663	0.755	–	–
I	0.522	0.518	0.713	0.693	0.685	0.768	–
B	0.424	0.433	0.540	0.562	0.609	0.661	0.707

Note: S = Symbolic Value; ES = Aesthetic Value; A = Attitude; SN = Subjective Norm; BC = Behavioural Control; I = Purchase Intention; B = Behaviour. The en dash (–) indicates duplicate correlation values omitted from the upper triangular portion because the matrix is symmetric.

Evaluation using heterotrait–monotrait ratio (HTMT) yielded a value of 0.919 for Attitude and Subjective Norm construct pair. This value slightly exceeds the conservative threshold of 0.90 recommended by Henseler et al. [46]. However, several methodological studies state that HTMT values of up to 0.95 are still acceptable in social science research when the constructs being tested have strong conceptual closeness and are supported by an adequate theoretical foundation [47]. In addition, the results of construct reliability, convergent validity, and the Fornell–Larcker criterion showed adequate results. Therefore, both constructs are retained in the model because they have an important theoretical role in explaining the Intention and Behaviour of batik use among Gen Z.

As additional empirical evidence regarding construct differences, a chi-square difference test was also conducted within the Covariance-based structural equation modelling (CB-SEM) model. The test was conducted by comparing the unconstrained and constrained models, namely by constraining the correlation between Attitude and Subjective Norm to one. The unconstrained model produced a value of $\chi^2 = 650.220$ ($df = 357$), while the constrained model produced a value of $\chi^2 = 835.386$ ($df = 358$). The difference between the two models resulted in $\Delta\chi^2 = 185.166$ with $\Delta df = 1$ ($p < 0.001$), indicating that forcing A and SN into one construct significantly decreased the model fit. These results indicate that A and SN remain two empirically distinct constructs. Therefore, the relatively high HTMT value reflects the conceptual closeness of the two constructs within the TPB framework rather than a failure of discriminant validity. These findings reinforce the Fornell–Larcker criterion results and support a more comprehensive evaluation of discriminant validity. This approach aligns with the recommendations of Anderson and Gerbing [48] and Kline [49], who suggest using the Chi-square Difference Test as additional evidence to confirm construct differences in covariance-based SEM.

Although the HTMT value between Attitude and Subjective Norm (0.919) slightly exceeded the conservative threshold of 0.90, several pieces of evidence support the discriminant validity of these constructs. First, both constructs represent theoretically distinct dimensions within the TPB. Second, the Fornell–Larcker criterion indicated that the square root of AVE for Subjective Norm (0.806) remained higher than its correlation with Attitude (0.798). Third, all indicators loaded highest on their intended constructs, and potentially overlapping items were removed during the measurement refinement process. Additionally, the structural model highlighted the distinct behavioural roles of the two constructs, where Attitude significantly affected Intention, while Subjective Norm failed to do so. This indicates that the observed HTMT value represents conceptual closeness and not construct redundancy. Therefore, the two constructs were retained in the final model. Table 6 shows the 4 highest HTMT values for construct pairs.

Table 6. Highest heterotrait–monotrait ratio (HTMT) values

Construct Pair	HTMT	Note
S $\leftrightarrow$ ES	0.821	Valid
A $\leftrightarrow$ SN	0.919	Marginal
A $\leftrightarrow$ I	0.820	Valid
BC $\leftrightarrow$ I	0.812	Valid

Note: S = Symbolic Value; ES = Aesthetic Value; A = Attitude; SN = Subjective Norm; BC = Behavioural Control; I = Purchase Intention.

Multicollinearity testing using the VIF in Table 4 showed that all VIF values were below the maximum threshold of 5. According to Hair et al. [43], a VIF value below 5 indicates that the model does not experience multicollinearity, which could disrupt the stability of parameter estimates. Therefore, the relationships between the independent variables in the model remain within acceptable limits, and the model is suitable for further SEM structural analysis.

4.2.2 Goodness of fit test

SEM analysis using AMOS 24 software provided information on the level of fit between the constructed structural model and empirical data through a number of goodness of fit (GOF) indicators. The evaluation of the suitability of the model was carried out on the best model as shown in Table 7, with reference to three main categories, namely absolute fit, incremental fit, and parsimony fit.

Table 7. GOF for the SEM model

GOF Category	Measure	Model Output	Acceptable Value	Decision
Absolute fit	SRMR	0.035	$\leq$0.08	Fit
	RMSEA	0.064	$\leq$0.08	Fit
Incremental fit	IFI	0.924	$\geq$0.90	Fit
	NFI	0.847	$\geq$0.90	Modest
	TLI	0.911	$\geq$0.90	Fit
Parsimony fit	PGFI	0.670	$\geq$0.50	Fit
	PNFI	0.739	$\geq$0.50	Fit
Normality	Multivariate critical ratio	2.601	$\leq$2.58	Modest

Note: GOF = goodness of fit; SEM = structural equation modelling; SRMR = standardised root mean square residual; RMSEA = root mean square error of approximation; IFI = incremental fit index; NFI = normed fit index; TLI = Tucker–Lewis index; PGFI = parsimony goodness-of-fit index; PNFI = parsimony normed fit index.

The results of the absolute fit test show that the structural model has acceptable model fit. The standardised root mean square residual (SRMR) value was 0.035 and the root mean square error of approximation (RMSEA) was 0.064. Both are below the maximum limit of 0.08. These values indicate that the model is able to represent the data well and meet the good fit criteria.

Furthermore, Also, Incremental fit testing was used to compare the proposed model against the null model. This test resulted in an incremental fit index (IFI) value of 0.924, a normed fit index (NFI) value of 0.847, and a Tucker-Lewis index (TLI) value of 0.911. Values for IFI and TLI greater than 0.90 suggest a good fit while NFI is within the moderate fit category. Even though most literature suggests values above 0.95 for excellent fit, the obtained values are appropriate enough for models with more than one latent construct [49, 50]. Thus, these results suggest that the model has satisfied the acceptable fit criteria, although it is not at the optimal level of fit yet.

With respect to parsimony fit, the parsimony goodness of fit index (PGFI) score of 0.670 and the parsimony normed fit index (PNFI) of 0.739 suggest that the model is quite simple in its structure but does not lose its ability to be explained. In other words, these scores exceed the suggested criteria, PGFI $\geq$ 0.5 and PNFI $\geq$ 0.5, which means that the model is parsimonious and structurally feasible [51]. This test also proves that most of the indices have passed into the fit category including RMSEA, IFI, and TLI. Even though the NFI score belongs to the modest category, such a result is quite normal due to the complexity of the model and the number of latent constructs.

This fit model adds validity to the use of the proposed model as a decision-making support system. The reason behind this is that since the model fits the empirical data, the connections between Symbolic Value, Aesthetic Value, Attitude, Behavioural Control, Intention, and Behaviour may be used as a structured decision-making priority. For the practical part, the fit model allows batik MSMEs to decide what product features need to be improved, made simpler, re-positioned, or distributed better in order to boost adoption by Gen Z consumer segment. Table 7 is GOF for the SEM model.

4.2.3 Hypothesis testing

Table 8 presents the path coefficients of the direct and indirect effects in the structural model.

• **Hypothesis H1–H3: Symbolic Value**

Table 8 shows that Symbolic Value influences Purchase Intention via the mediating TPB variables in different ways. First, Symbolic Value has an indirect significant effect on intention through Attitude (S $\rightarrow$ A $\rightarrow$ I; critical ratio = 1.98; p = 0.048). The finding implies that Symbolic Value enhances intention by forming positive consumer evaluation of batik as a product. Second, Symbolic Value has no indirect significant effect on Intention through Subjective Norm (S $\rightarrow$ SN $\rightarrow$ I; critical ratio = 0.968; p = 0.333). The finding implies that although Symbolic Value influences social legitimacy, it does not enhance intention by social norms. Finally, Symbolic Value has an indirect significant effect on Intention through Behavioural Control (S $\rightarrow$ BC $\rightarrow$ I; critical ratio = 2.595; p = 0.009). The finding implies that Symbolic Value enhances Purchase Intention by increasing consumers' belief in their capability to wear and use batik.

Table 8. Path coefficient of direct and indirect effects

Path	Estimate	SE	Critical Ratio	p-Value	Decision
			Direct Effect		
S → A	3.077	1.811	2.370	0.018	Supported
S → SN	2.046	1.299	2.397	0.017	Supported
ES → A	-2.312	1.299	-1.829	0.068	Rejected
ES → SN	-1.348	0.924	-1.634	0.104	Rejected
S → BC	1.677	0.591	4.053	***	Supported
ES → BC	-0.994	0.409	-2.560	0.011	Not supported
SN → I	0.139	0.120	1.059	0.336	Rejected
A → I	0.419	0.129	2.684	0.004	Supported
BC → I	0.361	0.098	3.366	0.001	Supported
I → B	0.845	0.101	7.284	***	Supported
			Indirect Effect		
S → A → I	0.485	0.675	1.98	0.048	Supported
S → SN → I	0.395	0.408	0.968	0.333	Rejected
S → BC → I	0.793	0.306	2.595	0.009	Supported
ES → A → I	-0.824	0.545	-1.512	0.131	Rejected
ES → SN → I	-0.192	0.216	-0.888	0.374	Rejected
ES → BC → I	-0.346	0.170	-1.87	0.062	Rejected
A → I → B	0.255	0.101	2.523	0.012	Supported
SN → I → B	0.093	0.089	1.047	0.295	Rejected
BC → I → B	0.243	0.079	3.063	0.002	Supported

Note: SE = standard error; p-value = probability value; S = Symbolic Value; ES = Aesthetic Value; A = Attitude; SN = Subjective Norm; BC = Behavioural Control; I = Purchase Intention; B = Behaviour. The arrow (→) indicates the direction of the hypothesised relationship. A relationship was considered statistically significant when $p < 0.05$. *** indicates $p < 0.001$.

These conclusions are supported by the results concerning the direct effects of the constructs. Symbolic value has a significant positive effect on Attitude (Estimate = 3.077, critical ratio = 2.370, p = 0.018), confirming H1. It implies that Gen Z evaluates batik not only as an item of clothing but also as a product that is meaningful in terms of identity and cultural significance. A positive Attitude arises from the perception that batik corresponds to consumers' individual expression and cultural needs. It confirms previous findings stating that the consumption of fashion items is a way of self-expression and identity construction among young consumers [19].

Symbolic Value also has a significant positive influence on Subjective Norm (S → SN; Estimate = 3.114; SE = 1.299; critical ratio = 2.397; p = 0.017), confirming H2. The findings confirm that symbolic meaning can contribute to consumers' perception of social legitimacy and appropriateness. Nevertheless, since the mediation from Subjective Norm to Intention is insignificant, social approval alone does not drive Purchase Intentions of Gen Z. In other words, Symbolic Value is more effective in its role as personal meaning.

Besides, Symbolic Value has a significant positive effect on Behavioural Control (S → BC; Estimate = 2.397; SE = 0.591; critical ratio = 4.053; $p < 0.001$), confirming H3. The finding implies that symbolically meaningful batik increases consumers' beliefs in their ability to buy, wear and use it. In other words, the symbolic meaning of a product not only forms positive Attitudes but also eliminates barriers to adoption by creating more psychological ease and contextuality [52].

From the managerial perspective, the results mean that Symbolic Value should be considered as one of the key attributes of the product. Batik MSMEs should transform Symbolic Value of their products into strategic product and communication decisions through motif stories, heritage stories, images of local identity, branding, packaging and brand authenticity. Simultaneously, Symbolic Value should be connected to usability-oriented product decisions, such as daily wearability, campus usage, casual events and semi-formal occasions. In other words, Symbolic Value should promote both emotional involvement and actual usage of the products.

• **Hypothesis H4–H6: Aesthetic**

Aesthetic Value does not exert a statistically significant indirect effect on batik Purchase Intention via the mediators of TPB. As seen from Table 8, the indirect effects via the mediating variables such as Attitude (ES → A → I, p = 0.131), Subjective Norm (ES → SN → I, p = 0.374), and Behavioural Control (ES → BC → I, p = 0.062) are insignificant. Thus, it can be concluded that Aesthetic Value does not positively affect Purchase Intention through the mediators of the study model.

In addition, the results of the direct effect analysis indicate that Aesthetic Value does not significantly influence Attitude (ES → A, p = 0.068), so H4 is rejected. Hence, one may conclude that Gen Z does not consider aesthetics as the major determinant of their Attitude towards batik. According to studies, Gen Z considers the Symbolic Value, Cultural Value, and relevance of the product to their own identity rather than just its visual attractiveness [53].

However, from the perspective of product management, this does not mean that aesthetics is irrelevant. In other words, it means that aesthetic features should be considered along with usability features and the correspondence of the products with lifestyle. In such a way, a visually attractive batik product can fail to promote adoption if it looks too formal and complicated in styling and wearing. Consequently, aesthetic decisions should focus on the simplicity of the design, adaptability of colors, silhouette and fit according to Gen Z preferences.

Moreover, there is a significant and negative relationship between Aesthetic Value and Behavioural Control. Aesthetic Value has a significant negative effect on Behavioural Control (Estimate = -0.994, critical ratio = -2.560, p = 0.011). Because the hypothesised relationship was positive, H5 was not supported and the significant relationship occurred in the opposite direction. Thus, H5 is not supported since it supposed a positive effect of Aesthetic Value on Behavioural Control. This result implies that batik, which is highly attractive from an aesthetic point of view, could also be considered impractical to use and wear. Though it is still a tentative assumption, it implies that batik MSMEs should take into account wearability and usability of the product when assessing Aesthetic Value.

Finally, Aesthetic Value does not have any significant impact on Subjective Norm ($\mathrm{ES} \to \mathrm{SN}$, p = 0.102), so H6 is rejected. In other words, it can be concluded that aesthetic assessment is mainly subjective. Overall, these results imply that batik MSMEs should apply more diversified strategies in product development considering different batik designs such as minimalist, casual, modern, formal and culturally distinctive.

• Hypothesis H7: Subjective Norm

The findings regarding indirect influence are given in Table 8. No significant influence of Subjective Norm on Behaviour was noted. The path from Subjective Norm through Purchase Intention to Behaviour ($\mathrm{SN} \to \mathrm{I} \to \mathrm{B}$) has a critical ratio of 1.047 and p-value of 0.295 (>0.05). This means that Purchase Intention could not mediate influence of Subjective Norm on Behaviour. Direct influence of Subjective Norm on Purchase Intention ($\mathrm{SN} \to \mathrm{I}$) was also insignificant, with a critical ratio = 1.059 and p = 0.289 (>0.05). To put it differently, Behaviour formation was not significantly affected by social pressure and normative expectations. These findings provide a basis for rejecting hypothesis H7 and imply that Subjective Norm does not play any important role as a determinant of either Purchase Intention or Behaviour formation. From a theoretical perspective, it can be said that social pressure and normative expectations did not affect the Behaviour of Gen Z directly. It appears that decision-making related to wearing batik is more personal and autonomous [54] than normatively driven. This finding corresponds to fashion-related research where it was found that Subjective Norm does not always predict Purchase Intention [40].

• Hypothesis H8: Attitude

From the findings above, it can be seen that there is a significant positive influence of Attitude on Intention ($\mathrm{A} \to \mathrm{I}$, critical ratio = 2.684; p = 0.004) and the indirect relationship of Attitude to Behaviour via Intention ($\mathrm{A} \to \mathrm{I} \to \mathrm{B}$), which is significant as well (critical ratio = 2.523; p = 0.012). Therefore, the results prove the H8 and show that the Intention acts as a mediator for the relationship between a favourable Attitude towards batik and Behaviour in terms of using it. It can be concluded that Gen Z will be motivated to buy or use batik if they consider it relevant to their lifestyle or culture. From the standpoint of management, this means that the positive Attitude of consumers towards batik must be enhanced by certain actions of batik MSMEs.

• Hypothesis H9: Behavioural Control

The results show that Behavioural Control positively and significantly influences Purchase Intention ($\mathrm{BC} \to \mathrm{I}$, critical ratio = 3.366; p = 0.001) and the indirect effect of the relationship between Behavioural Control and Behaviour via Purchase Intention ($\mathrm{BC} \to \mathrm{I} \to \mathrm{B}$) is statistically significant (critical ratio = 3.063; p = 0.002). These findings are consistent with previous TPB studies demonstrating that Behavioural Control acts as an important predictor of Purchase Intention [40, 55]. In this case, the Purchase Intention of Gen Z to buy and use batik is significantly influenced by factors related to the affordability, accessibility, ease, and convenience of the product. If the consumer finds batik affordable, accessible, easy to use, and convenient to incorporate into their outfit, then the consumer is likely to perceive that they can easily purchase and use it. The determinants of Behavioural Control have two components: a functional component and a psychological component. These components include external facilitating factors such as price, availability, online availability, and self-efficacy in performing the task.

It means that Gen Z is more likely to have intentions to purchase or wear batik when the product is perceived as affordable, accessible, easy to obtain, and convenient to use. In this way, for batik micro-small and medium-sized enterprises, it is directly implied what aspects of product development should be taken into account. These aspects include price, availability, distribution channels, size and style range, and wearing comfort.

• Hypothesis H10: Intention

Based on the results from the direct effect analysis as shown in Table 8, Intention has a very significant impact on Behaviour. The relationship path from intention to Behaviour ($\mathrm{I} \to \mathrm{B}$) has a critical ratio value of 7.284 and is a very significant p-value (<0.001). It may be concluded that intention is a very important determinant in explaining actual behaviour. As seen here, Intention is important in promoting the use of batik behaviour. In this case, hypothesis

H10 is validated. This can be compared with previous research that Gen Z follows Intentions [34].

As far as management is concerned, intention may be regarded as a leading predictor of product adoption and market demand. The relationship between Intention and Behaviour suggests that when Gen Z has intentions to purchase or wear batik, it is highly likely that intention will be converted into consumption. In other words, MSMEs involved in batik production can make use of intention as a factor for decision-making in terms of demand prediction, new product introduction, product selection from the product line, and marketing campaign development.

4.2.4 R^2 and effect size evaluation

According to the analysis of coefficient of determination (R^2), it can be concluded that the research model possesses high predictive power for the explanation of the behaviour of batik use of Gen Z (Table 9). For the Intention construct, the value of R^2 was 0.72, which means that Attitude, Subjective Norm, and Behavioural Control variables are able to explain 72% of the variances in batik usage intention. As mentioned by Hair et al. [43], an R^2 equal to 0.75 is considered to possess strong explanatory power, 0.50 moderate, and 0.25 weak.

Table 9. R^2 and effect size evaluation

Endogenous Variable	Predictor	R^2	f^2	Effect Size Interpretation
Attitude	Symbolic Value, Aesthetic Value	0.70	0.36	Large
Subjective Norm	Symbolic Value, Aesthetic Value	0.64	0.15	Moderate
Behavioural Control	Symbolic Value, Aesthetic Value	0.41	0.16	Moderate
Intention	Attitude, Subjective Norm, Behavioural Control	0.72	0.20	Moderate–Large
Behaviour	Intention	0.86	0.51	Large

The Behaviour construct showed the highest R^2 value equal to 0.86. Such values prove that the Intention variable has very strong predictive power for explaining the actual behaviour of batik use by Gen Z.

Additionally, the constructs for Attitude and Subjective Norm also exhibited high levels of explanatory ability as indicated by R^2 values of 0.70 and 0.64 respectively. This suggests that the constructs for Symbolic Value and Aesthetic Value can sufficiently explain a considerable amount of respondent evaluations of Attitude and social norms concerning batik use. However, the R^2 value of Behavioural Control was found to be 0.41, implying that the model has moderate explanatory power.

The effect size (f^2) evaluation results indicate that the relationship between Intention and Behaviour has the largest influence, with an f^2 value of 0.51, which is categorised as a large effect based on Cohen's criteria. The predictor variables of Attitude account for much of the explanatory power in the model. Nonetheless, the overall explanatory power in the model needs to be separated from the significance of each structural path in the model. It is observed that the effect of Symbolic Value is statistically significant while the effect of Aesthetic Value on the dependent variable (Attitude) is not statistically significant. Thus, the strong explanatory power of the model with regard to the dependent variable does not imply that both predictors have statistically significant effects individually. Overall, based on the R^2 and effect size values, the model has high explanatory power.

4.3 Policy Implication

The results of this study provide implications for policy and management. These implications are relevant for batik product design and marketing strategies among Gen Z. All implications are derived from the findings of direct and indirect influences between variables in the research model.

4.3.1 General implications for batik product management

The results show that symbolic value is an important determinant of both attitude, behavioural control, indirectly, and intention. That implies that Gen Z perceives batik not only as a fashionable product but also as a cultural product [56, 57]. For batik MSMEs, this finding implies that the management of symbolic value is a strategic priority. Consequently, for batik MSMEs, symbolic value must not only be promoted but must also be reflected in product strategy. Managers of batik MSMEs need to turn symbolic meaning into tangible product characteristics in terms of motif, collections, naming, package stories, and even digital stories. This will ensure not only preservation but also repositioning of batik as a culturally meaningful and relevant product for Gen Z consumers. However, the implications of this study may be different for various batik MSMEs. For heritage-oriented batik MSMEs, they preserve and emphasise motif authenticity, cultural stories, and history in their product design. Fashion-oriented batik MSMEs should redefine their symbolism in terms of contemporary design solutions. At the same time, batik

MSMEs that focus on the lifestyle market must embed the cultural identity of their products in consumers' everyday experience through identity branding and stories.

4.3.2 Management implications in batik product design

The conclusions drawn imply that aesthetic value has a significant negative effect on behavioural control. In turn, the revealed correlations between aesthetic value and behavioural control are substantially negative. It means that higher visual attractiveness does not always contribute to Gen Z consumers' ability to select, coordinate, buy, and wear batik. Visually overloaded designs, excessive ornamental patterns, rigid silhouettes, and a strongly ceremonial character may increase the effort necessary for integrating batik into one's daily wardrobe. As a result, even though consumers can recognize the aesthetic features of a batik item, they can perceive it as difficult to style, inconvenient to wear in informal settings, or incongruous with their wardrobe [58].

First, batik MSMEs need to make designs meant for younger people simpler. It means making them less dense, better balanced color-wise, and less formal. However, it does not mean stripping batik of cultural value and essence. It just implies the adaptation of traditional batik designs, such as using smaller designs, minimalist layouts, and partial batik [59, 60]. Second, product lines should be diversified. Since traditional batik often consist mainly of ceremonial shirts, batik MSMEs need to develop other product lines, such as casual shirts, outerwear, blouses, scarves, tote bags, bucket hats, and other modern clothes [59, 61]. Third, batik MSMEs need to invest in their digital channels. Since the purchasing process for younger consumers relies not only on product fit, but also on ease of finding, understanding, buying and receiving the product, having a digital catalogue, convenient purchasing and delivery options can help overcome obstacles and attract Gen Z consumers [62].

Since Gen Z is known to have low purchasing power, batik MSMEs need to consider developing tiered product lines—starting from cheaper products and going up to more expensive ones with greater complexity and symbolism of the design. The research has revealed that for successful adoption by Gen Z, batik needs to combine aesthetic relevance and usability.

The findings suggest that aesthetic value does not directly imply improved behavioural control. Thus, merely being aesthetically attractive is insufficient in order to motivate Gen Z to using batik. Visual attractiveness may create psychological impediments in case the products are seen as too formal, visually complex or challenging to coordinate with one's everyday look despite their attractiveness [58]. In other words, the designs of batik items should take into consideration not only attractiveness but also functionality, convenience and styling versatility.

4.3.3 Management implications in batik marketing strategy

Based on the results, subjective norm does not have a significant influence on Intention, showing that the effect of social pressure and normative expectations on batik adoption among Gen Z is not considerable. It means that appeals to obligations or conformity have less power in marketing campaigns than appeals to the relevance of products, individual self-expression, and convenience.

At the same time, attitude and behavioural control positively affect the adoption of batik. Therefore, Gen Z is likely to prefer batik that they find personally relevant and convenient to wear. This way, marketing activities need to improve positive attitudes by making batik look like a representation of personality and by improving behavioural control by highlighting ease of use, styling possibilities, and convenience of purchase [63].

Indirect effects show that symbolic value affects Intention via attitude and behavioural control, indicating that the impact of cultural meanings is stronger when people internalize them. Hence, storytelling appears to be a critical marketing tool. Batik MSMEs need to communicate the meanings of motifs, the stories of artisans and cultures, in order to make batik relevant to the current lifestyle of youths [64].

Digital media play a crucial role in it. As digital natives, Gen Z reacts positively to narrative, visual, and interactive content. This way, batik marketing must include not only pictures of products but also short videos, advice on styling, artisan stories, user-generated content, information about prices, and even direct links for purchases. Digital communication can reduce information barriers and increase behavioural control in this case.

To conclude, batik marketing among Gen Z must focus on meaning and lifestyle relevance more than on social obligation. Cultural heritage-oriented MSMEs can appeal to cultural pride and the philosophy of motifs; fashion-oriented businesses can highlight the creativity and flexibility of batik clothes; and lifestyle-oriented firms can present batik as a part of the current lifestyle.

4.3.4 Policy implications for the batik industry

These results imply the necessity for more comprehensive policy and industry support. As long as subjective norm does not have any effect on intention, policies aimed only at ceremonial use of batik will hardly yield tangible results. Therefore, the supporting measures should promote batik as an effective, wearable, affordable, and marketable product for young customers.

Firstly, MSMEs should be supported with regard to innovation and development of managerial skills. This includes mentorship in design, cooperation with younger designers, acquisition of the knowledge related to the process of product development, and preparation for data-based decisions [65, 66]. This support may allow MSMEs to convert consumer insights into better product portfolio planning, design adaptations, pricing, and channel management.

Secondly, policy interventions should encourage product diversification. Batik MSMEs should be encouraged to expand their product lines beyond formal wear to include casual clothes, semi-formal clothes, accessories, and affordable products for young customers.

Thirdly, the increase in efficiency of pricing and production processes should be achieved through joint production, sourcing, cost-efficiency training, and financing options. These activities will make it possible to offer affordable products without loss in quality.

Fourthly, the policy should support digitalisation and market access. MSMEs should get help with entering e-commerce markets, creating digital catalogues, taking product photos, posting content on social networks and creating videos about products due to the fact that Gen Z is learning and evaluating the products online. The policy should encourage intergenerational cooperation between artisans, young designers, universities, digital platforms, and local governments. Such a collaboration may help in translating traditional cultural values into products that is culturally significant, adaptable and relevant to the modern market [67].

In terms of engineering management, policy support should develop operational competencies such as product portfolio development, standardisation of design, quality control, digital inventory management, analysis of customer feedback, and small-batch innovation testing. In general, policy strategy should depend on the typology of MSMEs, i.e., heritage-oriented MSMEs should be supported in preserving culture and expanding the market, fashion-oriented MSMEs should be assisted in redesigning products and improving their functionality, and lifestyle-oriented MSMEs should receive help in brand-building and identity-based innovation.

4.3.5 Product management framework for batik micro, small, and medium enterprises

In order to enhance the value of engineering management within the scope of the study, the SEM results can be used as input to a product management framework for batik MSMEs. In addition to being analysed solely for consumer behaviour, the results can be used as input for the management of product design, product portfolio, market positioning, pricing, and channel configuration (Table 10).

Table 10. Product management framework derived from SEM findings

SEM Finding	Product Management Dimension	Actionable Implication for Batik MSMEs
Symbolic value significantly influences attitude and behavioural control.	Product design adjustment	Embed cultural meaning into motif selection, collection themes, product naming, packaging narratives, and storytelling.
Aesthetic value has limited influence and may reduce behavioural control.	Design simplification	Reduce motif density, use adaptive color combinations, simplify silhouettes, and improve mix-and-match usability.
Attitude and behavioural control significantly influence purchase intention.	Portfolio planning	Develop product lines for daily casual use, campus/office wear, semi-formal wear, and limited cultural collections.
Subjective norms do not significantly influence purchase intention.	Market positioning	Shift from obligation-based positioning to identity-based, lifestyle-relevant, and self-expression-oriented positioning.
Purchase intention positively influences behaviour.	Channel configuration	Use social media, e-commerce, short videos, styling tutorials, and user-generated content to demonstrate everyday use.
Gen Z consumers have relatively limited purchasing power.	Pricing strategy	Offer entry-level, middle-range, and premium product tiers to balance affordability, quality, and cultural value.

Note: SEM = structural equation modelling; MSMEs = micro, small, and medium enterprises; Gen Z = Generation Z.

Firstly, since there is a significant impact of symbolic value on attitude and behavioural control, it shows that the cultural significance of batik needs to be included in the product design process. There are several ways to translate symbolic value into the product development process, including choice of motifs, theme of collections, product naming, packaging stories, and even storytelling online.

Secondly, since the influence of aesthetic value is small and negative, care should be taken to manage visual aesthetics. Batik design that is too complicated, formal, and difficult to wear together with other clothes can decrease the level of ease of use. As a consequence, MSMEs should apply design simplification principles, namely, lower motif density, adaptive colors, silhouettes, and better compatibility of the products.

Third, since there are significant effects of attitude and behavioural control on purchase intention, product portfolio planning should be clearer and more structured. The portfolio of batik can be created according to the context of usage and can consist of several types of batik: daily casual batik, campus and office batik, semi-formal batik, and limited cultural collections.

Fourth, the insignificant role of subjective norms implies that market positioning should not be mainly based on the principle of cultural obligation and social pressure. On the contrary, batik should be positioned as a relevant and meaningful product for Gen Z.

Lastly, the positive influence of purchase intention on behaviour indicates that the channel setting should minimize barriers that prevent consumers from translating their purchase intention into actual behaviour. Batik enterprises can use online channels like social media, ecommerce websites, short videos, styling videos, and content created by consumers to showcase the wearability of batik in everyday situations. Such channels can highlight product usability.

Ultimately, the pricing strategy should be aligned with the purchasing power of Gen Z. Batik enterprises can provide various products ranging from low-end products and medium-range products to high-end products. Such tiered products can include simple designs, products suitable for daily use and products with high craftsmanship or narrative values.

5 Conclusion

This research analysed purchase intention of Gen Z towards batik through TPB augmented with symbolic value and aesthetic value. Based on the results, it can be concluded that symbolic value has the greatest impact on the TPB model, as it significantly affects attitude and behavioural control of Gen Z towards batik. Therefore, it can be stated that Gen Z recognizes batik as not only fashionable item but heritage product as well.

In terms of the results of SEM analysis, some of the hypotheses were supported. Symbolic value significantly affected attitude and behavioural control. Also, attitude and behavioural control significantly impacted purchase intention, whereas purchase intention significantly affected the behaviour. Nevertheless, subjective norms were found to have no significant impact on purchase intention, which means that neither social pressure nor cultural obligation plays a role in the adoption of batik among Gen Z. Moreover, the aesthetic value did not significantly affect attitude and negatively affected behavioural control.

In the context of engineering management and heritage-fashion product management, the key finding is the condition for batik adoption by Gen Z in terms of the capacity of MSMEs to transform symbolic meanings into products that are functional, affordable and contextualized. Consequently, batik MSMEs must not only retain the traditional motifs, but also design more product lines such as daily casual batik, campus or office batik, semi-formal batik and limited cultural collections. The design process must also involve design simplification, which includes simple composition of motifs, flexible color combinations and silhouettes which are more easily mixed and matched.

From the perspective of management, the results of the study give a straightforward message: it is time to rethink batik—not only as a formal product, but also as a relevant and versatile heritage fashion product for everyday life. The product strategy must take into account several aspects at once, including greater usage opportunities, affordable pricing for young people, online sales channels, readily available styling tips and meaningful stories about cultural values for Gen Z lifestyles.

In this research, there is a specific contribution in the sense that it reveals the incorporation of symbolic and aesthetic values into TPB to provide an explanation for purchase intention and behaviour of Gen Z regarding batik. Further research needs to consider measuring batik consumption through strict parameters like purchase and wearing frequency, occasion of use, and expenditure.

Author Contributions

Conceptualization, I.D.W.; methodology, D.A.P.; validation, S.M.A.; formal analysis, I.D.W.; investigation, I.D.W.; resources, S.M.A.; data curation, D.A.P.; writing—original draft preparation, I.D.W.; writing—review and editing, D.A.P.; visualization, S.M.A.; supervision, I.D.W. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflicts of interest.

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